

1.9. Build a Simple MVVM Application Using KnockoutJS (Local Setup)

Purpose:

To understand the Model-View-ViewModel (MVVM) pattern by building a simple application using KnockoutJS with manual file setup (without relying on a CDN). This will demonstrate how observables and computed observables work in a reactive UI.

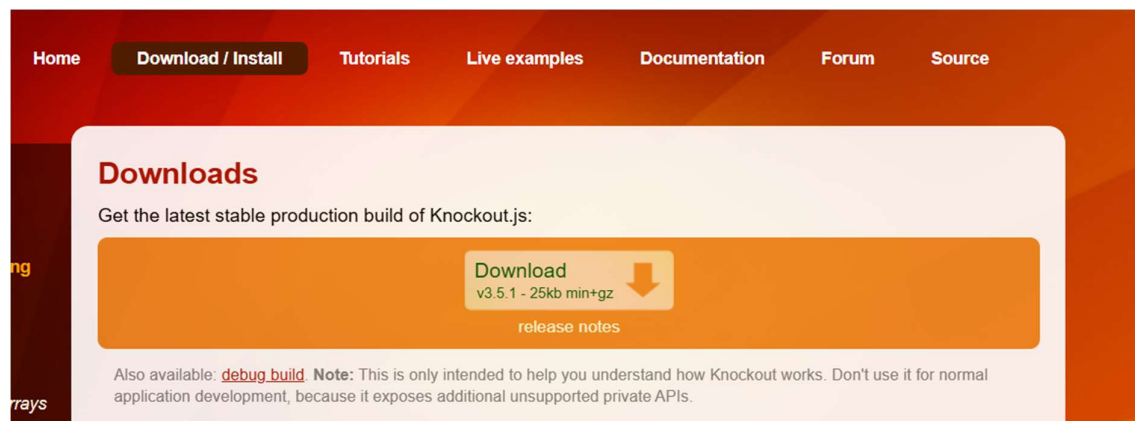
Prerequisites:

- Basic knowledge of HTML and JavaScript
- Familiarity with the MVVM pattern (Model-View-ViewModel)
- A browser (e.g., Chrome, Firefox)
- A code editor (e.g., VS Code, Notepad++)
- KnockoutJS library file (downloaded manually)

Step 1: Download KnockoutJS Library

1. Visit the official KnockoutJS website:

<https://knockoutjs.com/downloads/index.html>



2. When you click in download, the following screen will open

```
← → ↺ 🌐 knockoutjs.com/downloads/knockout-3.5.1.js

/*!
 * Knockout JavaScript library v3.5.1
 * (c) The Knockout.js team - http://knockoutjs.com/
 * License: MIT (http://www.opensource.org/licenses/mit-license.php)
 */

(function() {
    (function(n) {
        var A=this||{0:eval}("this"),w=A.document,R=A.navigator,v=A.jQuery,H=A.JSON,v||"undefined"===typeo
        define(["exports","require"],n:"object"===typeof exports&&"object"===typeof module?n(module.exports|exports):n(A.ko={}))
        {var d;return function(){d||(d=a.a.setTimeout(function(){d=n;b()},c))}}function Y(b,c){var d;return function(){clearTimeout(
        d=a.a.setTimeout(b,c))}function Z(a,c){c&&"change"!==c?"beforeChange"===c?this.pc(a):this.gb(a,c):this.qc(a)}function aa(a,c
        (this.Qb&&this.mb[c]?d.uc(c,a,this.mb[c]),this.mb[c]=null,--this.Qb):e.I[c]||d.uc(c,a,e.J)?{da:a:d.$c(a)},a.Ja&&a.gd())}var
        d=b.split("."),e=a,f=0;f<d.length-1;f++)e=e[d[f]];e[d[d.length-1]]=c;a.L=function(a,c,d){a[c]=d;a.version="3.5.1";a.b("ver
        a.version");a.options={deferUpdates:!1,useOnlyNativeEvents:!1,foreachHidesDestroyed:!1};a.a=function(){function b(a,b){for(va
        (a[c]=b[c]);return a}function d(a,b){a.__proto__=b;return a}function e(b,c,d,e){var l=b[c].match(q)||[];a.a.D(d.match(q),fun
        {__proto__:[]})instanceof Array,h="function"===typeof Symbol,m={},k={};m[R&&/Firefox\/2/i.test(R.userAgent)?
        "KeyboardEvent":"UIEvents"]=["keyup","keydown","keypress"];m.MouseEvents="click dblclick mousedown mouseup mousemove mouseov
        c=0,d=b.length;c<d;c++)k[b[c]]=a}};var l={propertychange:!0},p=w&&function(){for(var a=3,b=w.createElement("div"),c=b.getEle
        \x3e",c[0]);return 4<a?a:n)}(),q=/\S+/g,t;return {jc:["authenticity_token",/^__RequestVerificationToken(.*?)$/],
        D:function(a,b,c){for(var d=0,e=a.length;d<e;d++)b.call(c,a[d],d,a)},A:"function"===typeof Array.prototype.indexOf?function(a
        c=0,d=a.length;c<d;c++)if(a[c]==b)return c;return-1},lb:function(a,b,c){for(var d=0,e=a.length;d<e;d++)if(b.call(c,a[d],d,a
        b.splice(d,1):0==d&&b.shift()),wc:function(b){var c=[];b&&a.a.D(b,function(b){0>a.a.A(c,b)&&c.push(b)});return c},Mb:functi
        b c\}var d=1;if(a)for(var e=0,l=a.length;a[l-e-1]e++d push(h call(c a[l-e] all) return d) if(h function(a b c\}var d=1;if(a)for(va
```

3. Save this file to your project directory (choose **"Save Link As..."**), where your actual project files saved.



Step 2: Create Project Files

Create a new folder, e.g., knockout-mvvm-demo, and inside that folder, create:

- index.html
- app.js
- knockout.js (already downloaded)

Step 3: Add HTML (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Knockout MVVM Local Example</title>
</head>
<body>

  <h1>Knockout MVVM Example</h1>

  <p>First name: <input data-bind="value: firstName" /></p>
  <p>Last name: <input data-bind="value: lastName" /></p>
```

```
<h2>Hello, <span data-bind="text: fullName"></span>!</h2>

<script src="knockout-3.5.1.js"></script>

<script src="viewModel.js"></script>

</body>
</html>
```

Step 4: Add JavaScript (viewModel.js)

```
function AppViewModel() {
  this.firstName = ko.observable("Richa");
  this.lastName = ko.observable("Arora");

  this.fullName = ko.computed(() => {
    return this.firstName() + " " + this.lastName();
  });
}

window.addEventListener('DOMContentLoaded', function () {
  ko.applyBindings(new AppViewModel());
});
```

Step 5: Run the Application

1. Open the index.html file in any modern browser.
2. Try changing the First name and Last name fields.
3. You'll see the full name update automatically in real-time.

Knockout MVVM Example

First name:

Last name:

Hello, Richa jain!

What You Learned:

- Setting up KnockoutJS locally
- Creating observables using `ko.observable`
- Creating computed observables using `ko.computed`
- Binding values in HTML using `data-bind`
- Applying MVVM in a simple JavaScript application